

Neil Varshney

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Education

Bachelor of Computer Science Co-op
Carleton University

Ottawa, ON
Sept 2023 – Present

- **Field:** Computer & Internet Security
- **CGPA:** 10.88/12 (A-)
- **Awards:** Dean's List (2x), Chalmers Jack MacKenzie Scholarship, Henry Marshall Tory Scholarship

Technical Skills

Languages & Frameworks: Python, Java, C/C++, JavaScript/TypeScript, SQL, HTML/CSS, React, Flask, FastAPI

Tools & Technologies: PostgreSQL, MongoDB, SQLite, Cypher, Polars, PyTorch, Pandas, RESTful APIs, JWT, Git, Linux, Power Automate, SharePoint

Experience

Data Engineer & Threat Detection Analyst Co-op
Employment & Social Development Canada (ESDC)

Gatineau, QC
May 2026 - Present

- Built manual **ETL pipelines** in **Python** using **Polars** to clean Parquet files and build a **LadybugDB** graph database
- Modeled **HTTP request logs** in **LadybugDB** to extract session data and map account relationships for a **fraud detection platform**
- Analyzed program datasets to **detect high-risk activity** and presented statistical threat findings to management

Data Management & Digital Transformation Co-op
Health Canada

Ottawa, ON
Jan 2026 – Apr 2026

- Created a comprehensive **Data Catalog** and **Data Dictionaries** on **SharePoint** to standardize the organization of scientific data assets
- Designed **SharePoint** site structures and implemented **metadata tagging systems** to streamline collaborative workflows and file management
- Developed and implemented automated workflows using **Power Automate** to streamline **data entry** and notification processes

Projects

Fake News Detector || *ML, Python, JavaScript, React, SQLite, HTML/CSS*

 [Fake-News-Detector](#) 

- Trained a machine learning classifier using BERT embeddings and logistic regression to detect fake news, reaching **98.8% accuracy** on a dataset of **8,000+ articles**
- Built a **Python REST API** backed by **SQLite**, using **JWTs** and password hashing to **secure user authentication** and protect analysis endpoints
- Developed a **React** frontend that displays real-time model **confidence scores** and logs user query history

Extracurricular Activities

Carleton Computer Science Society: Member
Intermural Soccer: Participant

Sept 2023 – Present
Sept 2024 – Present